

1. PixelProof: AI image detection

A working prototype, measured improvements
and the limits of generalization

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2. Two tasks, two kinds of evidence

Model1: Does this image contain AI-generation evidence?

False alert: a real photograph receives an AI warning.

Miss: an AI image escapes detection.

Model2: Where might a local AI edit be?

A region map needs its own ground truth and controls.

3. Strong early scores did not transfer

Table 1 shows why a familiar test score was only a starting point.

Table 1. Early historical accuracy comparisons.

Model	Familiar data	External collection
Small CNN	96.75%	77.1%
ResNet-18	97.66%	25.2%

Table 1 motivated controls for resolution, source and labels.
Later audits also qualified the external benchmark.

4. Measurement had to be repaired

Reversed labels

Explicit source mapping and reruns

Evaluation-driven threshold

Calibration-only selection

Training learns parameters. Calibration chooses the rule.
Development reveals weaknesses. A final test stays independent.

These roles must remain separate even when files differ.

5. The system combines existing representations

Pretrained visual features

DINOv2 · CLIP · DEAR

Project adaptation

Audited data · fitted components · constrained correction

Validated upload → E92 scores → evidence and uncertainty

6. Real-image false alerts fell substantially

Figure 1 compares the same 160 real images per condition.

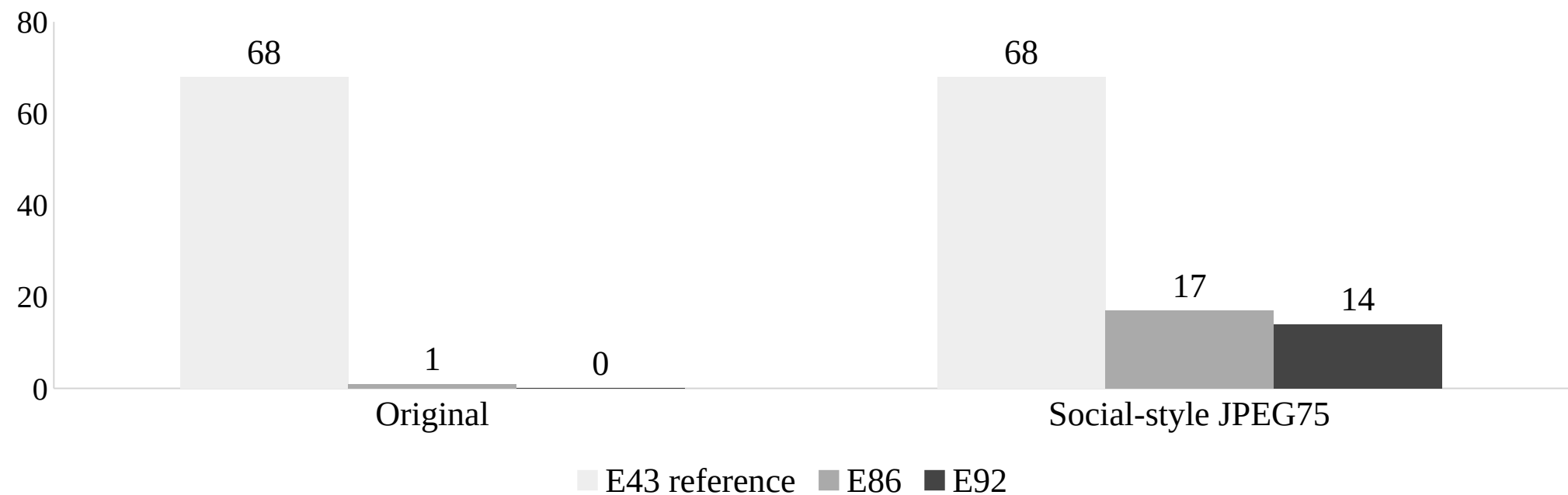


Figure 1. False alerts out of 160 real development images.

Reused development data · E92 AI detection: 159/160 in each view

7. 20/20 checks did not mean full acceptance

Table 2 separates numeric checks from full acceptance.

Table 2. Two distinct acceptance questions.

Question	E92 outcome
Numerical development criteria	20/20 passed
No newly missed E43-detected AI image	Failed: one original AI image

Table 2 explains the limit: better totals can hide a new error.
The checks were correlated and were not 20 independent datasets.

8. The displayed score is not a probability

Table 3 uses score cuts, not authenticity probabilities.

Table 3. Simplified E92 display rules.

Evidence	User-facing result
Original score $\geq$ upper cut (~ 7.94)	AI evidence; review warning if needed
Both views $<$ lower cut (~ 1.15)	No clear AI evidence
Remaining cases	Uncertain

Table 3 does not certify authenticity. E43 is advisory.
Decisions use unrounded scores; 50% is not the boundary.

9. Broader checks exposed remaining weaknesses

Owner gallery: 206 unique real-image files

False AI alerts: 9 originals; 18 social-style views

Previously inspected, mostly one phone and correlated scenes

Later research candidates were rejected

Source-fold evaluation failed the complete requirements.

New source-pixel features harmed transported AI recall.

10. Model2 remains an experimental pilot

Table 4 compares edit placement on 16 consumed parents.

Table 4. Localization trade-off, original-image condition.

Measure	Earlier head	Two-placement head
Original-placement pixel AUC	0.742	0.699
New-placement pixel AUC	0.566	0.638
Authentic falsely flagged area	17.11%	25.74%

One editor · adaptive development · candidate rejected

11. A traceable prototype was delivered

Local web demonstration

Validated uploads · clear uncertainty · guarded inference

Reproducible engineering

Pinned artifacts · experiment receipts · dataset provenance
Git history · automated software checks

Latest recorded checkpoint: 1,245 Python tests passed.

12. What the project achieved—and did not

Table 5 separates implementation success from scientific proof.

Table 5. Evidence-supported outcome.

Achieved	Still unproven
Working student research demo	Universal detection
Fewer development false alerts	Independent final-test success
Auditable corrections and rejected candidates	Calibrated authenticity probability
Controlled localization experiments	Reliable AI-edit localization

A credible student prototype with clearly stated limits.

13. The next step is stronger evidence

Independent sources and resolved ancestry

Separate calibration, fixed costs and locked evaluation

Broader cameras, generators, editors and processing

My central lesson: verify what a score actually proves.

14. Selected references: detection and data

Wang et al. (2020)

CNN-generated images are surprisingly easy to spot... for now. CVPR.

Ojha et al. (2023)

Towards universal fake image detectors. CVPR.

Grommelt et al. (2024)

Fake or JPEG? arXiv:2403.17608.

Abdelhamed et al. (2018)

A high-quality denoising dataset for smartphone cameras. CVPR.

Chen et al. (2018)

Learning to see in the dark. CVPR.

Keleş (2026)

PixelProof research records, source commit 497d45c. GitHub.

15. Selected references: features and evaluation

Radford et al. (2021)

Learning transferable visual models from natural language supervision. PMLR.

Oquab et al. (2024)

DINOv2: Learning robust visual features without supervision. TMLR.

Kim et al. (2026)

Dissect and prune: Enhancing robustness in AI-generated image detection. ICML.

Guo et al. (2017)

On calibration of modern neural networks. PMLR.

Dwork et al. (2015)

Generalization in adaptive data analysis and holdout reuse. NeurIPS.

Türk Telekom (2026a, 2026b)

2025 Annual Report; corporate About Us page.